

Drone Design and Aerodynamic Analysis Project Report

Project Summary

This project presents the conceptualization, modeling, and performance evaluation of a compact drone developed by **KONDRU CHARWICK HAMESH**. The core aim was to enhance aerodynamic efficiency, validate structural stability, and analyze the dynamic interaction of mechanical components during flight. A detailed 3D model of the drone was created, incorporating key elements such as the frame, impeller blades, and gear mechanisms. These components were then subjected to simulation-driven refinement to ensure optimal functionality.

Through iterative design and testing, the project explored how each subsystem contributes to the drone's overall flight stability and airflow efficiency. The integration of mechanical motion studies and fluid dynamics simulations provided a holistic understanding of the drone's behavior under operational conditions. This comprehensive approach—from CAD modeling to advanced simulation—demonstrates a robust methodology for modern drone development.

Software Tools and Workflow

The entire project was executed using the **SolidWorks** platform, leveraging its suite of design and simulation modules to streamline the development process:

- **SolidWorksCAD**
Used for precise 3D modeling of individual drone components, including the impeller, gear assemblies, and structural frame. These parts were assembled into a unified model (Assembly.SLDASM) to visualize and validate mechanical integration.
- **SolidWorksMotionStudy**
Enabled simulation of mechanical movements within the drone. Motors and constraints were applied to replicate real-world dynamics, such as blade rotation and gear interaction. This module also facilitated animation of the assembly for visual inspection and performance analysis.
- **SolidWorksFlowSimulation(CFD)**
Applied to assess aerodynamic characteristics. The tool simulated airflow patterns around the drone's propellers and body, revealing pressure zones, velocity vectors, and flow trajectories. These insights were critical for optimizing the drone's aerodynamic profile and ensuring efficient thrust generation.

By utilizing SolidWorks for both design and analysis, the project maintained a cohesive workflow, allowing for seamless transitions between modeling, motion simulation, and fluid dynamics evaluation.

Assembly Overview

The drone's complete structure was assembled in SolidWorks using the Assembly.SLDASM file, which integrates multiple custom-designed components into a cohesive and functional model. Each part was individually modeled and strategically positioned using mating constraints to define spatial relationships and enable realistic mechanical interactions.

Key Components in the Assembly

- **ImpellerBlades(1-ImpellerBlades.SLDPRT)**

These aerodynamic blades serve as the primary propulsion system, designed to generate lift by accelerating airflow. In the assembly, they are mounted onto a central hub and powered by the motor through a gear or belt transmission. Their geometry was optimized for thrust efficiency and aerodynamic performance.

- **ArmGear(2-ArmGear.SLDPRT)**

This gear interfaces directly with the rotor hub and motor drive system. It transmits rotational energy from the central gear to the impeller blades, ensuring synchronized motion. Accurate alignment and meshing within the assembly were essential for smooth and reliable power delivery.

- **GearingSystem(3-Gearing.SLDPRT)**

Acting as the primary transmission unit, this gear or pulley mechanism is affixed to the main body and driven by the motor. It connects to the arm gear via a belt or gear train, regulating the rotational speed and torque of the impeller blades. Gear ratios were considered to fine-tune performance parameters such as RPM and thrust.

- **MainStructure(5-MainStructure.SLDPRT)**

Serving as the central chassis, this component houses the motor and provides mounting points for all other subsystems. It supports the impeller assembly, gears, landing legs, and camera. Designed for both mechanical strength and aerodynamic efficiency, it anchors the entire assembly and ensures structural coherence.

- **LandingLegs(4-Legs.SLDPRT)**

Four stabilizing legs are attached to the underside of the main frame to facilitate safe takeoff and landing. Their placement ensures balanced support and sufficient clearance for the rotating impellers. The legs were symmetrically positioned and designed to absorb impact while maintaining ground stability.

- **CameraModule(6-Camera.SLDPRT)**

A fixed camera unit is mounted on the drone's body, typically at the front or bottom. It represents a payload or sensing system, integrated without compromising aerodynamic flow or structural balance. While not mounted on a

gimbal in this version, its placement demonstrates how auxiliary equipment can be incorporated into the drone design.

Assembly Integration

All components were assembled using precise mating techniques:

- **Concentric mates** for shafts and rotational parts
- **Coincident mates** for mounting surfaces
- **Gear and belt mates** to simulate mechanical transmission

The motor initiates rotation in the gearing system, which in turn drives the impeller blades via the arm gear. The legs provide ground support, and the camera remains fixed to the frame. This realistic assembly configuration was essential for conducting both motion simulations and aerodynamic (CFD) analyses.

Motion Study – Dynamic Simulation

To evaluate the mechanical behavior of the drone under operational conditions, a dynamic motion study was conducted using **SolidWorks Motion**. This simulation aimed to replicate the real-world movement of the propulsion system and assess the stability of interconnected components during rotation.

A virtual motor was configured to rotate the **Gearing System** at **100 RPM**, simulating the primary drive mechanism. This rotational motion was transmitted to the **Arm Gear** and subsequently to the **Impeller Blades** through either a **belt drive mate** or **gear mate**, mimicking the actual mechanical linkage found in drone propulsion systems. The simulation confirmed that the transmission path—from motor to gearing, through the arm gear, and finally to the impeller—operated smoothly without misalignment or slippage.

Additional angular controls were introduced to explore potential articulating features, such as swiveling arms or adjustable camera mounts. Although the primary motion focused on impeller rotation, the simulation also captured subtle structural responses, including vibrations or shifts in the **Main Structure** and **Landing Legs** due to the rotating mass.

The motion was animated over a continuous 360° rotation at the defined speed, producing a realistic visual output. The resulting animation (Motion.mp4) showcased the impeller blades spinning consistently while the rest of the drone remained stable. This visual validation demonstrated proper gear meshing, tensioned belt dynamics (if applicable), and the absence of collisions between moving parts and the drone's frame. The drone's stationary posture throughout the simulation indicated a well-balanced and mechanically sound design, ready for aerodynamic evaluation.

CFD Analysis – Aerodynamic Performance

Following the successful mechanical validation, a **Computational Fluid Dynamics**

(CFD) study was performed using **SolidWorks Flow Simulation** to analyze the drone's aerodynamic characteristics. The objective was to optimize airflow efficiency, evaluate pressure distribution, and understand the forces acting on the drone during flight.

The simulation modeled the drone in a hovering state, with **air** as the working fluid. The **Impeller Blades** were set to rotate at **150 RPM**, slightly higher than in the motion study, to intensify airflow and enhance the fidelity of aerodynamic analysis. Boundary conditions were defined to allow air intake near the impeller and exhaust through open surroundings, simulating an external flow domain. Solid boundaries were assigned to all drone surfaces, including the blades, body, and legs, to capture realistic fluid-structure interactions.

To simulate the effect of rotating blades without physically rotating the mesh, a **rotating region** or **moving reference frame** was employed. This approach allowed accurate modeling of induced airflow while maintaining computational efficiency.

Mesh and Solver Configuration

The flow domain was discretized into a mesh comprising **4,018 cells**, with **1,325 cells** directly interfacing with solid surfaces. This refinement ensured detailed resolution near critical areas such as blade edges and structural contours, capturing boundary layer effects and localized pressure gradients.

The CFD solver ran for **140 iterations**, with monitoring goals set for key parameters like velocity, pressure, and thrust. The simulation converged successfully, with all goals meeting predefined thresholds and no significant residual changes observed. The absence of warnings or errors confirmed the stability and robustness of the simulation setup.

This aerodynamic analysis provided valuable insights into the drone's thrust generation and airflow behavior, reinforcing the design's suitability for stable and efficient flight.

Parameter	Value
Status	Solver is finished.
Total cells	4,018
Fluid cells	4,018
Fluid cells contacting solids	1,325
Iterations	140
Last iteration finished	12:20:20
CPU time per last iteration	00:00:00
Travels	3.5
Iterations per 1 travel	40
CPU time	0:0:27
Calculation time left	0:0:0
Run at	PPATH\MESH\ASUS
Number of cores	16

Event	Iteration	Time
Mesh generation started	0	12:19:52, May 04
Mesh generation normally finished	0	12:19:53, May 04
Preparing data for calculation	0	12:19:53, May 04
Calculation started	0	12:19:54, May 04
Calculation has converged since the following criteria are satisfied:	140	12:20:20, May 04
Goals are converged	140	12:20:20, May 04
Calculation finished	140	12:20:20, May 04

Warning	Comment
No warnings	

CFD Solver Output Summary

The Computational Fluid Dynamics (CFD) simulation was executed efficiently, completing in just **27 seconds of CPU time** thanks to parallel processing on a **16-core system**. This rapid turnaround reflects both the manageable mesh size and the optimized performance of the SolidWorks Flow Simulation solver for this application.

The solver generated a mesh with **4,018 fluid cells**, and the simulation ran for **140 iterations**, successfully meeting all convergence goals. As shown in the solver log (Figure 2), key events such as mesh generation, solution initiation, and convergence confirmation were recorded. The message “Calculation has converged since the following criteria are satisfied: Goals are converged” followed by “Calculation finished” with no errors or warnings, indicates a stable and reliable simulation. These results validate the aerodynamic analysis and confirm that the data can be confidently used for performance interpretation.

Flow Trajectories and Aerodynamic Insights

With the simulation successfully converged, flow visualization tools were employed to analyze airflow behavior around the drone. **Streamlines** were generated in the Flow Simulation.mp4 video to depict the movement of air as influenced by the rotating impeller. These trajectories show air being drawn from above the drone into the impeller region and expelled downward, simulating thrust generation.

The streamlines are smooth and symmetric around the impeller axis, indicating a well-balanced propeller design that produces uniform airflow. The absence of significant turbulence or recirculation zones near the drone’s body and legs suggests minimal aerodynamic obstruction—an indicator of effective structural design.

In addition to streamlines, **surface pressure plots** were examined to assess pressure distribution across the drone’s components. The impeller blades exhibit a clear pressure gradient: lower pressure on the upper (suction) side and higher pressure on the lower

(pressure) side, which is responsible for lift. The **Main Structure** and other parts show mostly ambient pressure, with slight variations in areas directly exposed to airflow or in wake regions. For example, minor low-pressure zones appear near the impeller inlet, while the underside experiences pressure drops due to accelerated airflow—consistent with Bernoulli’s principle. The **Landing Legs** show increased pressure on wind-facing surfaces, confirming their exposure to aerodynamic forces.

These visualizations affirm that the impeller effectively generates thrust and that the drone’s structure is aerodynamically sound, with no extreme pressure concentrations that might compromise stability or integrity.

Comparing Motion and CFD Conditions

A notable distinction between the two simulations lies in the **impeller’s rotational speed**:

- **Motion Study:** 100 RPM
- **CFD Analysis:** 150 RPM

This discrepancy means the aerodynamic simulation reflects a more intense operating condition than the mechanical animation. In practice, a higher RPM results in increased airflow, greater pressure differentials, and enhanced lift. The choice to simulate at 150 RPM may represent a scaled-down approximation of real-world conditions or a deliberate test of the drone’s performance under elevated stress.

From a design perspective, this variation highlights the flexibility of simulation parameters and the importance of aligning mechanical and aerodynamic studies. For more accurate comparisons, future iterations could standardize the RPM across both simulations. Nevertheless, the qualitative outcomes remain consistent: the drone operates smoothly at 100 RPM mechanically, and performs effectively at 150 RPM aerodynamically.

Testing at a higher RPM in CFD also introduces a **safety margin**, suggesting that the drone would experience even less aerodynamic stress at lower speeds. This reinforces the robustness of the design across a range of operating conditions and provides valuable insight into how motor speed influences lift and stability.